

Table 1: Theorem 1 empirical check — MAMuJoCo (SafeDreamer, full comm). Does the bound Δ cover the actual gap between real and imagined cost?

Env	d	lag-lr	ϵ_c	TV (from KL)	Δ (bound)	Observed gap	Holds?	Slack	Seeds
HC 2x3	5.0	0.0001	0.0009 ± 0.0008	0.2227 ± 0.0126	23.4 ± 1.3	0.0161 ± 0.0155	✓	$\times 1451$	3
HC 2x3	5.0	1e-05	0.0009 ± 0.0004	0.2215 ± 0.0352	23.3 ± 3.7	0.0206 ± 0.0107	✓	$\times 1129$	5
HC 2x3	25.0	0.0001	0.0032 ± 0.0007	0.2005 ± 0.0234	21.1 ± 2.5	0.0477 ± 0.0166	✓	$\times 443$	3
HC 2x3	25.0	1e-05	0.0038 ± 0.0004	0.2270 ± 0.0294	23.9 ± 3.1	0.0844 ± 0.0714	✓	$\times 283$	3
Ant 2x4	0.2	0.0001	0.0009 ± 0.0006	0.4155 ± 0.0259	43.6 ± 2.7	0.0115 ± 0.0090	✓	$\times 3780$	3
Ant 2x4	0.2	1e-05	0.0004 ± 0.0001	0.3848 ± 0.0106	40.4 ± 1.1	0.0078 ± 0.0034	✓	$\times 5165$	3
Ant 2x4	25.0	0.0001	0.0052 ± 0.0003	0.3835 ± 0.0051	40.3 ± 0.5	0.1036 ± 0.0808	✓	$\times 389$	3
Ant 2x4	25.0	1e-05	0.0056 ± 0.0014	0.3886 ± 0.0264	40.9 ± 2.8	0.1446 ± 0.1202	✓	$\times 283$	3
Ant 4x2	1.0	0.0001	0.0005 ± 0.0002	0.3525 ± 0.0175	37.0 ± 1.8	0.0118 ± 0.0134	✓	$\times 3146$	3
Ant 4x2	1.0	1e-05	0.0004 ± 0.0001	0.3474 ± 0.0316	36.5 ± 3.3	0.0061 ± 0.0064	✓	$\times 5981$	3
Ant 4x2	25.0	0.0001	0.0034 ± 0.0003	0.3580 ± 0.0092	37.6 ± 1.0	0.0768 ± 0.0426	✓	$\times 490$	3
Ant 4x2	25.0	1e-05	0.0029 ± 0.0026	0.3776 ± 0.0196	39.7 ± 2.0	0.0354 ± 0.0369	✓	$\times 1121$	3

Formula:

$$\Delta = 15 \epsilon_c + 105 \cdot c_{\max} \cdot \text{TV}, \quad c_{\max} = 1 \text{ (binary velocity cost, exact from env code).}$$

What each column measures:

- ϵ_c — *Cost-head error*. How wrong the model prices each step’s cost. Source: average of `Model/cost_loss` over the last 10% of training rows.
- **TV** — *Transition error*. How wrong the model’s next-state forecast is. Source: average of `Model/div` (KL divergence, prior vs. posterior) over the last 10% of training rows. Converted to TV via Pinsker’s inequality: $\text{TV} = \sqrt{\text{KL}/2}$.
- **Δ (bound)** — The maximum gap Theorem 1 *allows* between real and imagined cost.
- **Observed gap** — The gap we *actually measured*: $|J_{\text{real}} - \hat{J}_{\text{imag}}|$.
Both sides scaled to a discounted 15-step window ($\times 13.9 = \sum_{t=0}^{14} \gamma^t$, $\gamma = 0.99$):
— Imagined: `Value/Cost` (per dreamed step) $\times 13.9$.
— Real: `main/cost` (episode total) $\div$ episode length $\times 13.9$.
Episode length = measured from consecutive `steps` diffs (not hardcoded).
- **Slack** — How much room to spare: $\Delta \div \text{gap}$.

Assumptions & caveats:

- **Average, not worst-case.** Theorem 1 requires uniform bounds on every state-action pair. All values here are training averages. This table is a consistency check (“does the data contradict the theorem?”), not a worst-case proof.
- **Tail-10% window.** We average each metric over the last 10% of logged rows, assuming the model has converged by that point.
- **Even cost spread.** Real per-step cost = episode total $\div$ episode length. This assumes cost is spread evenly, but MAMuJoCo cost is binary (0 or 1 per step) and likely bursty — risky windows may have much higher cost than the average suggests.
- **KL $\rightarrow$ TV conversion.** Pinsker gives a conservative upper bound on TV from KL, but both are computed on latent codes, not raw states. The theorem is stated over the true state space.